SANJAI S

Full Stack - Software Engineer

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Aspiring Full Stack - Software Engineer skilled in Java, Data Structures & Algorithms, SQL, and backend development. Strong foundation in object-oriented programming, problem-solving, and software development, with a focus on building efficient, scalable, and maintainable applications.

EDUCATION

BE.Computer Science Engineering (Artificial intelligence & Machine learning) 2023 - 2027

Velalar college of engineering & technology , Thindal, Erode. CGPA - 7.03 /10

TECHNICAL SKILLS

Programming Languages : Java, python, SQL.

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

Backend Development: Spring Boot, REST APIs.

Devops & Tools : Git & GitHub, Docker, Postman.

Cloud & Deployment: AWS, Render, Vercel.

Database : MySQL, Postgresql, supabase, Neon.

WORK EXPERIENCE

Software Development Intern — Tamil Nadu National Highways Department | Chennai, Tamil Nadu | Aug 2026 - Sep 2026
Awarded an Letter of Recommendation from the Deputy Chief Engineer.

Tamil Nadu National Highways Department Management System | Next.js, React, TypeScript, PostgreSQL, Supabase, Tailwind CSS

- Developed an enterprise full-stack web platform using Next.js, React, TypeScript, and PostgreSQL to manage highway corridors, infrastructure projects, maintenance contracts, legal cases, fee bills, and departmental staff.
- Designed and implemented REST API route handlers and relational PostgreSQL schemas across multiple production modules, maintaining strongly typed data models and 52 incremental database migrations.
- Implemented enterprise-grade authentication and authorization using TOTP-based 2FA, hierarchical RBAC, PBAC jurisdiction scoping, PostgreSQL Row-Level Security (RLS), and database-backed rate limiting.
- Built reusable progressive forms, dashboards, search/filtering systems, soft-delete workflows, and Excel export functionality across six core production modules.
- Developed an immutable audit trail capturing record creation, modification, deletion, restoration, and role changes, including field-level diffs and historical snapshots.

Inward Tapal Management System — Tamil Nadu Highways Department | Node.js, Express.js, PostgreSQL, JavaScript, Supabase

- Developed a full-stack inward letter and document tracking system to manage registration, assignment, workflow progression, and archival of official communications across departmental offices.
- Designed and implemented RESTful APIs using Node.js and Express.js for authentication, CRUD operations, search/filtering, workflow management, and system configuration.
- Implemented Role-Based Access Control (RBAC) for multiple administrative roles, along with date-lock mechanisms and authorization controls to prevent unauthorized modifications.
- Designed a PostgreSQL database schema with dedicated tables for users, offices, tapal records, workflow history, audit logs, and system settings, including triggers and constraints for data integrity.
- Built audit logging and workflow history to track user actions, status transitions, record updates, and security-related changes.
- Optimized database search using PostgreSQL pg_trgm text matching and connection pooling, with deployment support for Supabase, Vercel, and Linux servers using PM2.

PROJECTS

IPL Performance Analysis Dashboard | Python, Pandas, Streamlit, Plotly, Three.js

- Developed an interactive web application to process and analyze 301,000+ ball-by-ball deliveries across 1,169 IPL matches (2008–2025) using Python and Pandas.
- Engineered interactive 3D pitch maps and wagon-wheel visualizations using Three.js/WebGL, rendering up to 500 ball trajectories per view for detailed match analysis.
- Implemented reusable data-processing and analytical modules for phase-wise statistics, batter–bowler matchups, strike rates, and economy metrics.
- Optimized application performance for a 44.5 MB dataset using vectorized Pandas operations and Streamlit caching, reducing redundant computations and improving application responsiveness.
- Integrated data-processing, visualization, and interactive UI components into a single deployable application, providing a seamless end-to-end software solution.

CERTIFICATIONS

- **HackerRank SQL (Advanced)** Skill Certification – Demonstrated expertise in advanced SQL, including joins, CTEs, window functions, subqueries, aggregation, and query optimization. (July 2026)
- Deploying **Gen AI with AWS Workshop** – Sri Eshwar College of Engineering, Thiran 2025 (Feb 2025).

ACHIEVEMENTS

1st Prize – Smart Bus Optimization Technical Competition (2025).

Finalist - VIT Hackathon (2026).

Qualified in the College-Level Internal Hackathon for **Smart India Hackathon** (SIH) 2025.